Date:

To:

From: Mikey Wiesecke MEGR 3152 Section L09

Subject:

Tensile Test of 1018 Steel, 360 Brass, T6 Aluminum, and Cast Iron

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Objective

The objective of this lab was to learn how to use an Instron 5582 Universal testing machine to do tensile tests to compare the properties of three samples. Two types of tests were done in this experiment, one was a standard tensile test till failure and the other was a strain hardening test. With the data derived from this test we can make stress and strain graphs which allows us to analyze many types of mechanical properties of our samples. It's important to compare these results to vendors to show the effects of defects on mechanical properties.

Result and Discussion

Importance of the Extensometer

The Importance of an Extensometer in an accuracy of a stress and strain curve cannot be understated enough. Figure 1. Shows a comparison of two curves, one created using the extensometer and the other not using the extensometer.

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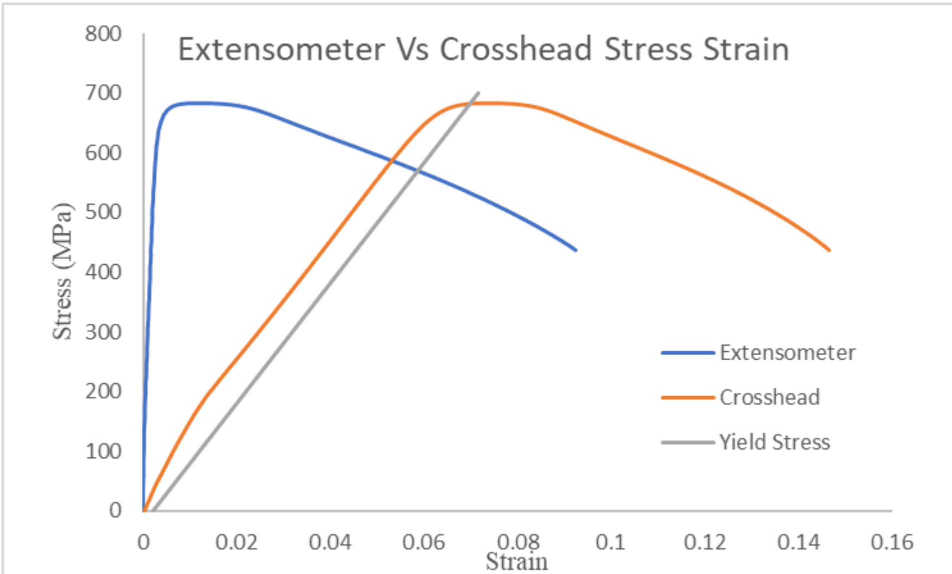

Figure 1. Extensometer Vs Crosshead Stress and Strain Graph

The two curves significantly differ with the crosshead displaying a material that is incredibly elastic and the extensometer displaying a more generic curve that is expected from steel. Of course, the crosshead curve is not completely wrong but it is inaccurate especially when analyzing the elastic region of the graph. Figure 2 shows a table comparing the properties of 1018 steel analyzed from both curves to the spec of a vendor.

Figure 1. Comparison Table between the Extensometer and Crosshead

	Usage of Extensometer	Young's Modulus (GPa)	Yield Strength	Ductility
Experimental Values	w/o	10.07	675.507	14.65
	w	461.33	634.82	9.24
Literature		200	440	15

From the table it can be concluded that the crosshead reported similar numbers for yield strength in addition to being closer to vendor specs in the category of Ductility. The deal breaker is young modules which was incredibly off being forty-five times lower than the extensometer and twenty times lower than the vendor spec. This is not great as the elastic region of the graph is where most material properties are gathered and analyzed.

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Mechanical Properties of 1018 Steel, T6 Aluminum, 360 Brass, and Cast Iron

By using the force and distance data we can develop engineering stress and strain graphs that we can use to analyze the mechanical properties of our samples. Figure 1 compares the values found by the test in comparison to the values from vendors and only sources.

Figure 1. Mechanical Properties Comparison

		Modulus of Elasticity (MPa)	Yield Strength (MPa)	Tensile Strength (MPa)	Ductility (%)	Resilience
Gray Iron	Experiment	136.86	266.6	346.53	0.8673244	0.666472
	Literature	62.1-1450	344.738	90-1650	.2-40	
1018 Steel	Experiment	461.33	634.82	682.43	9.2397109	1.33
	Literature	200	372.32	440	15	
T6 Aluminum	Experiment	86.16	271.73	344.9	14.900894	0.996
	Literature	68.9	275.79	310	17	
360 Brass	Experiment	66.4	282.58	389.7	28.897426	1.45
	Literature	97	310.264	338-469	53	

Starting with yield stress it can be observed that the experimental yield stress is less than the literature yield stress aside from the 1018 Steel which is an outlier. For yield stress this is too be expected as defects will often cause the strength of the material to lower. In addition, the structure of every material is not uniform, and one sample may provide different results from another. The strongest material would be 1018 Steel which is strange because it would be expected that Cast iron has the highest yield strength because it has significantly more carbon. This can be explained by the brittleness of Cast Iron, Cast Iron would have a greater yield strength but because Cast Iron is so brittle it can not be strained enough to reach its higher yield point. In short, the modulus of Cast Iron holds the yield strength back. Concluding the rest of data, the 1018 steel has the highest stiffness with a modulus of elasticity of 461.33, 360 Brass has the highest ductility with a percent of elongation of 28.9% and has the highest resilience. The following figures: fig 2., fig 3., fig 4., and fig 5., show the stress and strain curves of each material.

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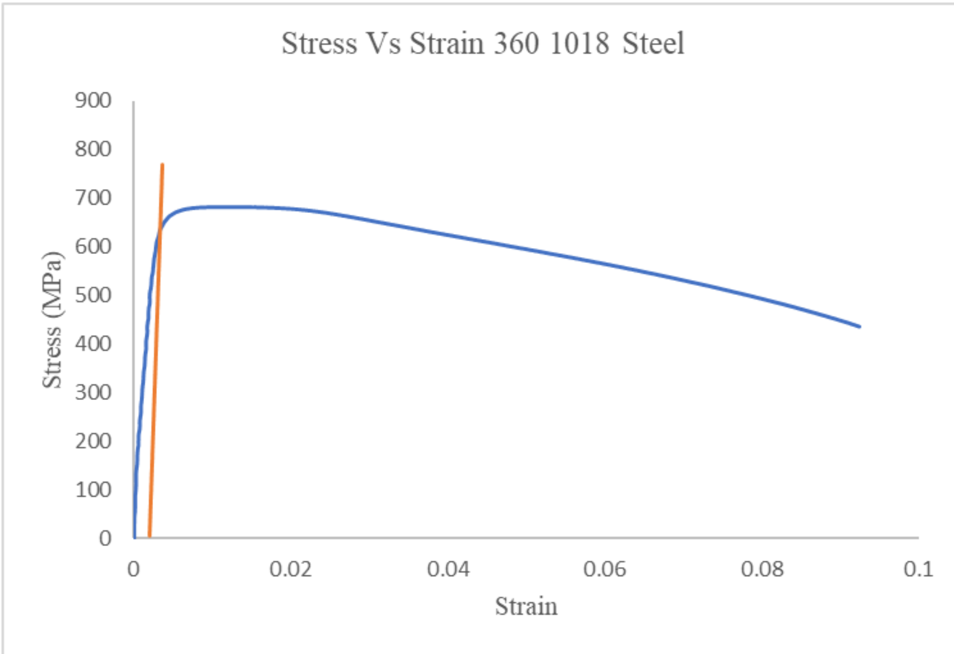

Figure 2. Stress Vs Strain Curve of 1018 Steel

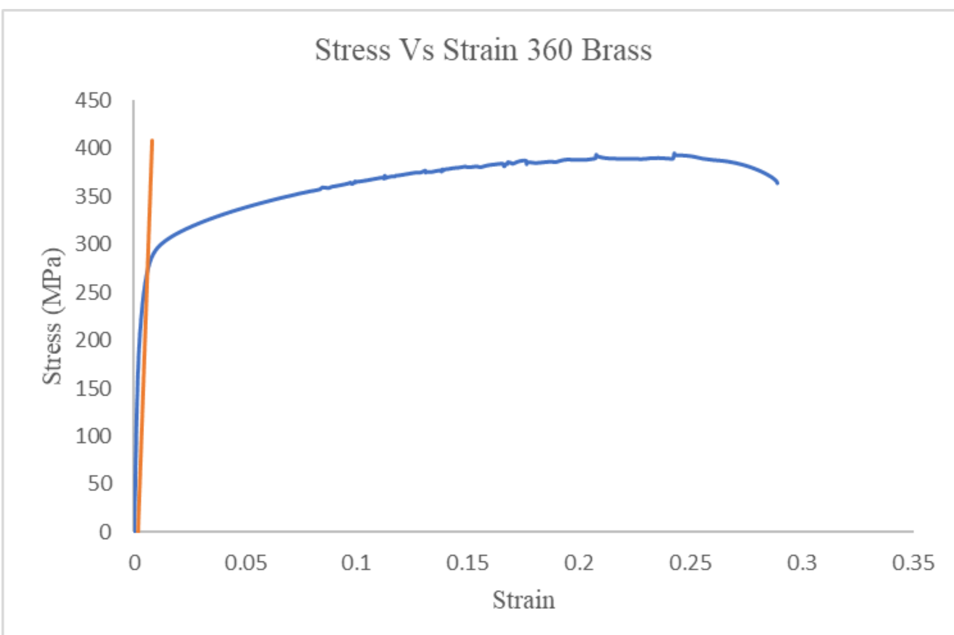

Figure 3. Stress Vs Strain Curve of 360 Brass

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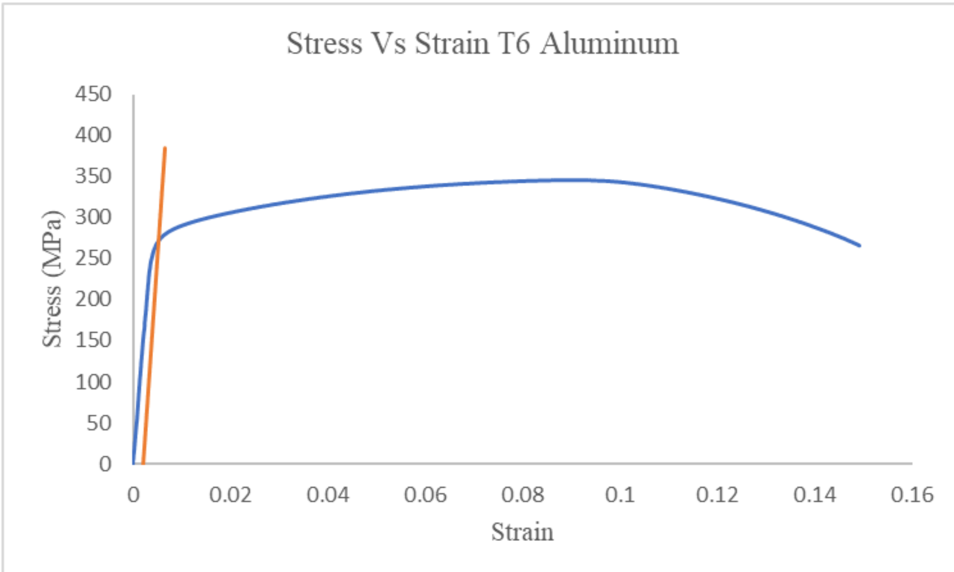

Figure 4. Stress Vs Strain Curve of Aluminum

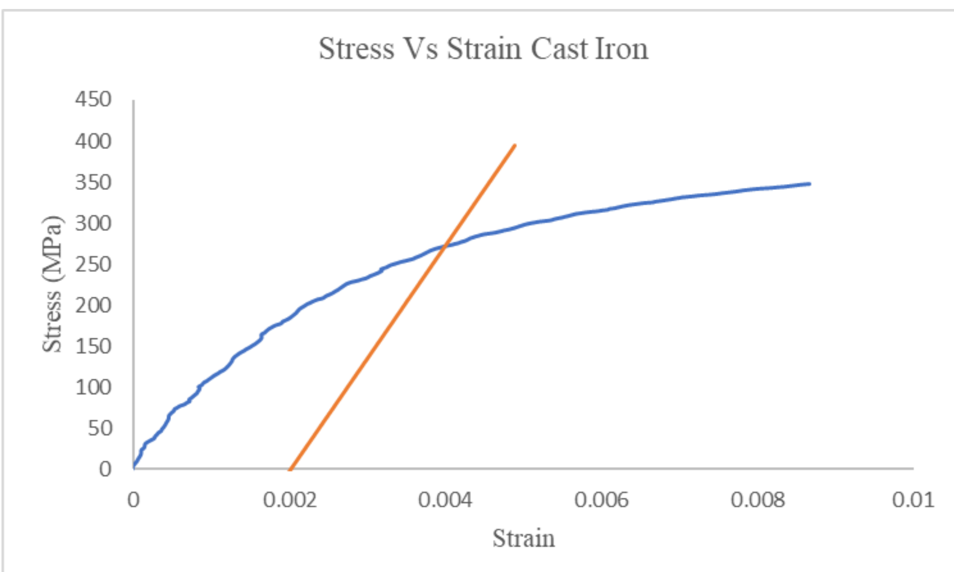

Figure 5. Stress Vs Strain Curve of Cast Iron

The yield point was found using the .002 strain offset rule which works for the linear Elastic section of the curve. The main outlier here being Cast Iron which did not show a linear elastic range so the yield stress may be inaccurate. Resilience was found by taking the area under the curve up to the yield point. This can be done by integrating the linear portion of the curve

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resulting in brass having the greatest about of resilience. 360 Brass has the greatest elastic ductility allowing it to absorb the most amount of energy without being plastically deformed.

True Stress and Strain Vs. Engineering Stress and Strain

The engineering stress and strain graph is a standard that is extremely useful when finding the properties of material and serves as a basic visual for deformation of a material under stress. However, the engineering stress and strain graph is not entirely true to what is happening to the material because the strain is measured as a difference from its starting value. For the actual stress strain curve the true stress and strain graph is created by calculating the change in distance of one point to the point previous and is iterated for every point. Figure 5 shows the difference between the true stress strain curve and the engineering stress strain curve.

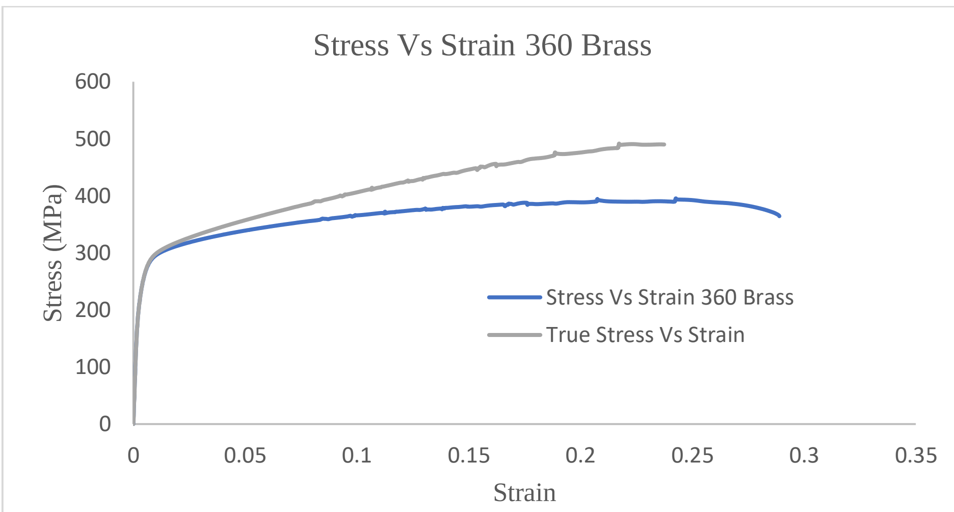

Fig. 5 True Stress/strain vs Engineering Stress/strain

The main difference between these two lines occurs after the yield point at which the true lines continue to strengthen while the engineering line settles at a values and eventually decreases before fracturing. The reason for this is because in real life a material will undergo strain hardening as it plastically deforms. The amount of strain hardening that a material may experience can be quantified with the strain hardening exponent (n) and can be found by graphing the natural log of both true stress and true strain and fitting the curve to a linear line. The slope of this line is going to be your strength hardening exponent (n). Figure 6. Shows the process being done to the same 360 brass as before.

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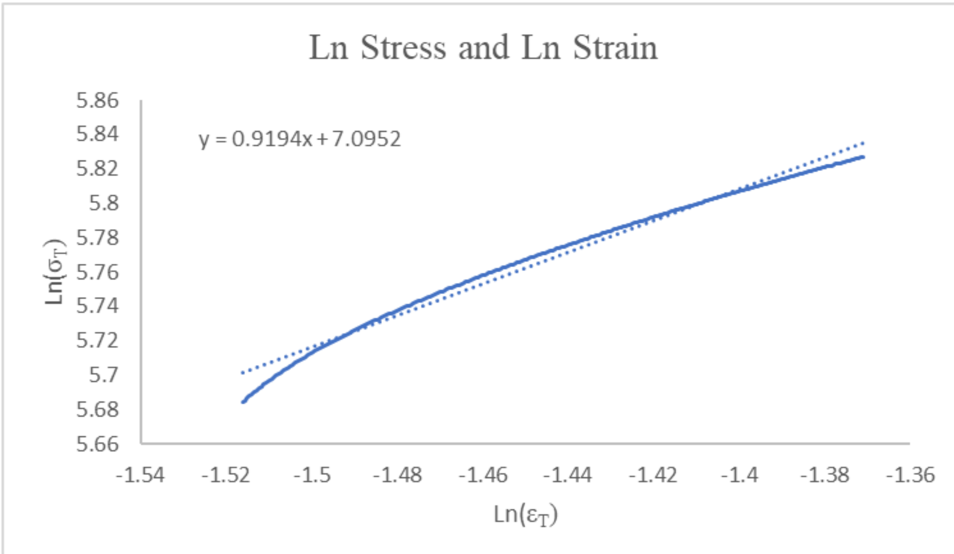

Fig.6 Natural log of True Stress vs Natural Log of True Strain of 360 Brass

Using Eq (1), the Hollomon Equation, and our hardening coefficient we can find the strength coefficient (k) of our material.

$$\sigma_T = k{\varepsilon_T}^n \qquad \text{Eq (1)}$$

Resulting in the strength coefficient being (k) 1201.83 MPa and strain hardening exponent (n) is 0.9194.

Cycled Strain Hardening of 1018 Steel

When thinking about strain hardening it's useful to graph and analyze a material under a cycle load. Ideally what happens in this case is the material will strain harden during its first cycle and will show a stress strain curve with a high modulus of elasticity and often a higher yield stress during its second cycle. Figure 7 shows this test done to 1018 Steel under two cycles.

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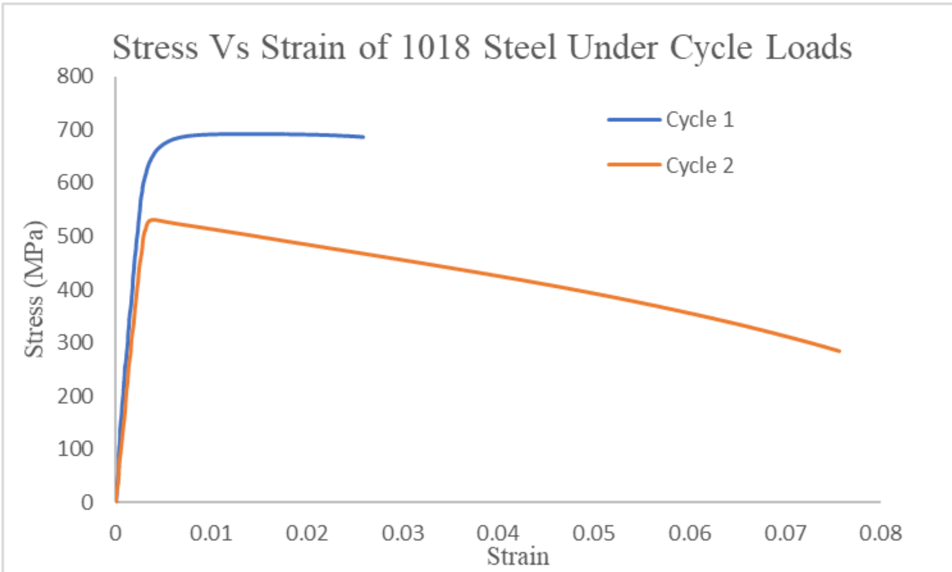

Fig. 7 Stress Vs Strain of 1018 Steel Under Cycle Loads

This test did not yield results that were to be expected. On the second cycle the young modulus of the steel did increase, which was to be expected. However, cycle one has a significantly higher yield stress which is not something that was expected. The main problem is that cycle two yield stress should be a lot higher or at least around that seven hundred number cycle one was at. Possibly the test was run for too long causing the sample to partially fracture during its first cycle leading to weaker results during cycle two. The sample looks to have almost undergone an annealing process bringing the yield stress down, but the increased stiffness still occurred. Truly these results are weird and aside from normal user error there seems to be no explanation of why the data looks the way that it does.

Conclusions

The Purpose of this experiment was to use tensile test to develop a stress and strain curves for different samples. These samples included 1018 steel, T6 Aluminum, 360 Brass and cast iron. It was concluded that out of these sample 1018 steel had the greatest yield stress while the 360 brass has the greatest resilience. The incompatibility of tensile test with cast irons were also examined. The 1018 steel that was examined during this test displayed properties much greater than that that was reported by online vendors and other sources. In addition to the basic properties that were examined using an engineering stress and strain graph, a true stress and true strain graph was developed to examine the effects of strain hardening. Using these graphs, it was also possible to find the strain hardening exponent of .9194 which leads to finding the strength coefficient of 1201 MPa using Holloman formula. To further observe the effects of strain hardening a test was run exposing the sample with tensile stress in cycles

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rather than just to failure. The test resulted in interesting results; the first cycle yielded a significantly higher yield stress compared to that of the second cycle. However, as expected the young modulus of the second cycle was higher than that of the first cycle. When the stress and strain curve were created from the data that was taken as an extension of the crosshead the young modulus was significantly lower than the more accurate model created from the extensometer data. Strangely the extensometer model proved to be less accurate in reporting the ductility than that of the crosshead model. This experiment showed the variability of mechanical properties and why proper tests are needed to ensure safety and functionality. This lab also showed the importance of knowing the effect of going beyond the elastic range of a material. These effects are very important to an engineer when designing a product or ensuring the safety of the user of said product.

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Appendices

Data Spreadsheet

Tensile Test 4 Metal Data Comparison